

FMCG ERP SYSTEM

# FMCG ERP Maintenance, Utilities & Factory Operations Manual

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Maintenance · Utility Management · ESG · IoT Monitoring

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**Audience:** Maintenance Technicians, Engineers, Factory Managers, Utility Officers

**Version:** v2.0 — Post-Recovery Edition

**Date:** 24 May 2026

**Status:** Confidential — Internal Use Only

ERP button/link recovery complete. All navigation targets restored and verified. Smoke test:  
141/141 routes passed. BVT: 0. Broken action cards: 0.

# FMCG ERP — Maintenance & Utilities Manual v2

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**Version:** 2.0 (post-recovery)

**Date:** 2026-05-24

**Audience:** Maintenance Managers, Engineers, Utility Managers, Factory Managers

**Modules Covered:** Maintenance · Assets · Breakdowns · Planned Maintenance · Predictive Maintenance · Spare Parts · Utilities (Electricity, Water)

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## 1. Module Overview

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**What it does:** Manages equipment and facility maintenance — asset register, breakdown logging, preventive maintenance scheduling, predictive maintenance alerts, spare parts inventory, and utility consumption tracking.

**Who uses it:**

- Maintenance Manager — schedules preventive maintenance, monitors asset health
- Maintenance Engineer — logs breakdowns, executes work orders

- Utility Manager — tracks electricity and water consumption
- Production Manager — coordinates planned downtime for maintenance

#### When to use it:

- When a machine breaks down and needs urgent repair
- When scheduling preventive maintenance (PM) activities
- When tracking spare parts usage and stock
- When monitoring electricity and water consumption
- When reviewing equipment uptime and MTBF

#### Modules at a glance:

| Feature            | Route                                                                     | Purpose                        |
|--------------------|---------------------------------------------------------------------------|--------------------------------|
| Maintenance        | <a href="/dashboard/maintenance">/dashboard/maintenance</a>               | Full maintenance workspace     |
| Utility Management | <a href="/dashboard/utility-management">/dashboard/utility-management</a> | Electricity and water tracking |

**Maintenance**  
Assets, breakdowns, planned maintenance, predictive, spares

**Overview** Assets Breakdowns Plans Predictive Spares Reports

**Maintenance**  
Asset register, preventive maintenance & breakdowns

**TOTAL ASSETS**  
0  
0 under maintenance

**OPEN BREAKDOWNS**  
0  
active / in-repair

**OVERDUE PMS**  
0  
past due date

**LOW STOCK SPARES**  
0  
at or below reorder level

**Open Breakdowns** View all  
No open breakdowns

**Overdue PM Plans** View all  
All PMs are up to date

**Asset Register**  
Manage machines, equipment and production lines

**PM Plans & Work Orders**  
Preventive maintenance schedules and completion tracking

**Breakdown Records**  
Log and resolve equipment failures with MES linkage

**Predictive Maintenance**  
IoT-based failure risk queue and maintenance recommendations

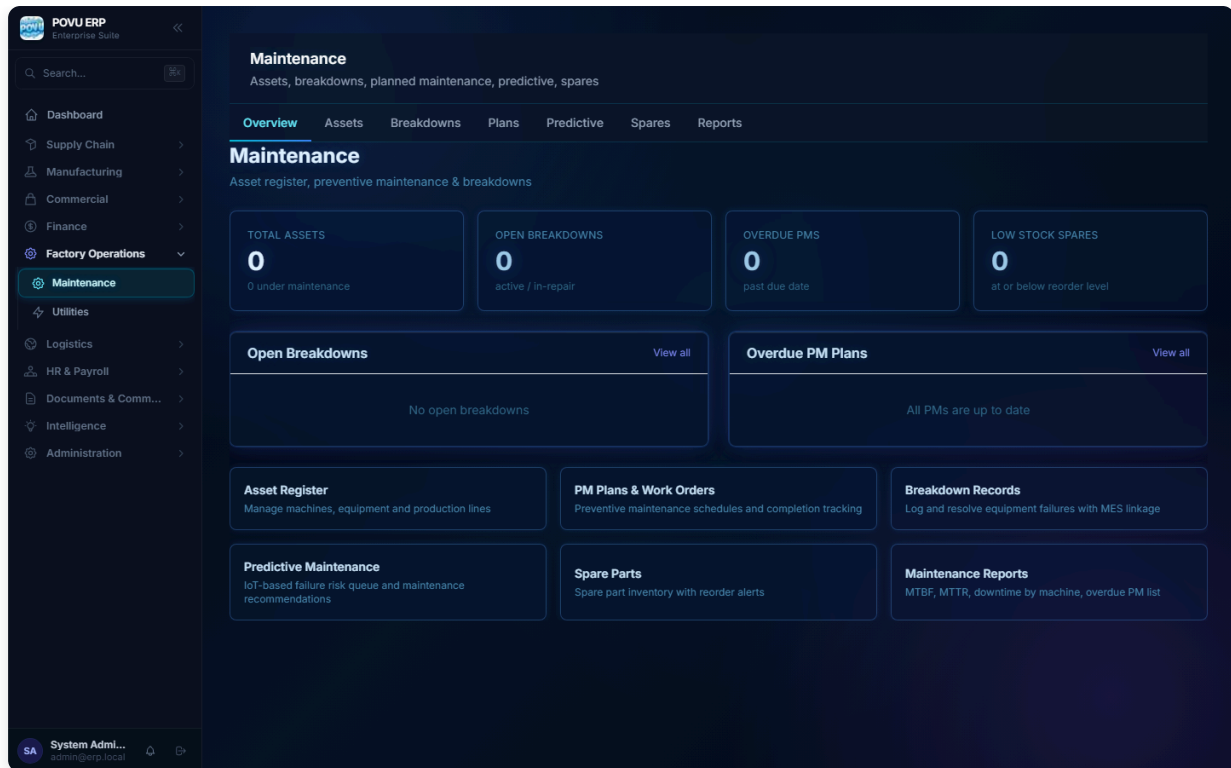
**Spare Parts**  
Spare part inventory with reorder alerts

**Maintenance Reports**  
MTBF, MTTR, downtime by machine, overdue PM list

*Maintenance module overview.*

## 2. Maintenance Overview Dashboard

Tab: Overview



Maintenance dashboard — assets count, open breakdowns, pending PM tasks, MTBF, MTTR.

KPI cards:

- Total Assets (active)
- Open Breakdowns
- Planned PM Tasks Due This Week
- Average MTBF (Mean Time Between Failures)
- Average MTTR (Mean Time To Repair)
- Spare Parts Low Stock Alerts

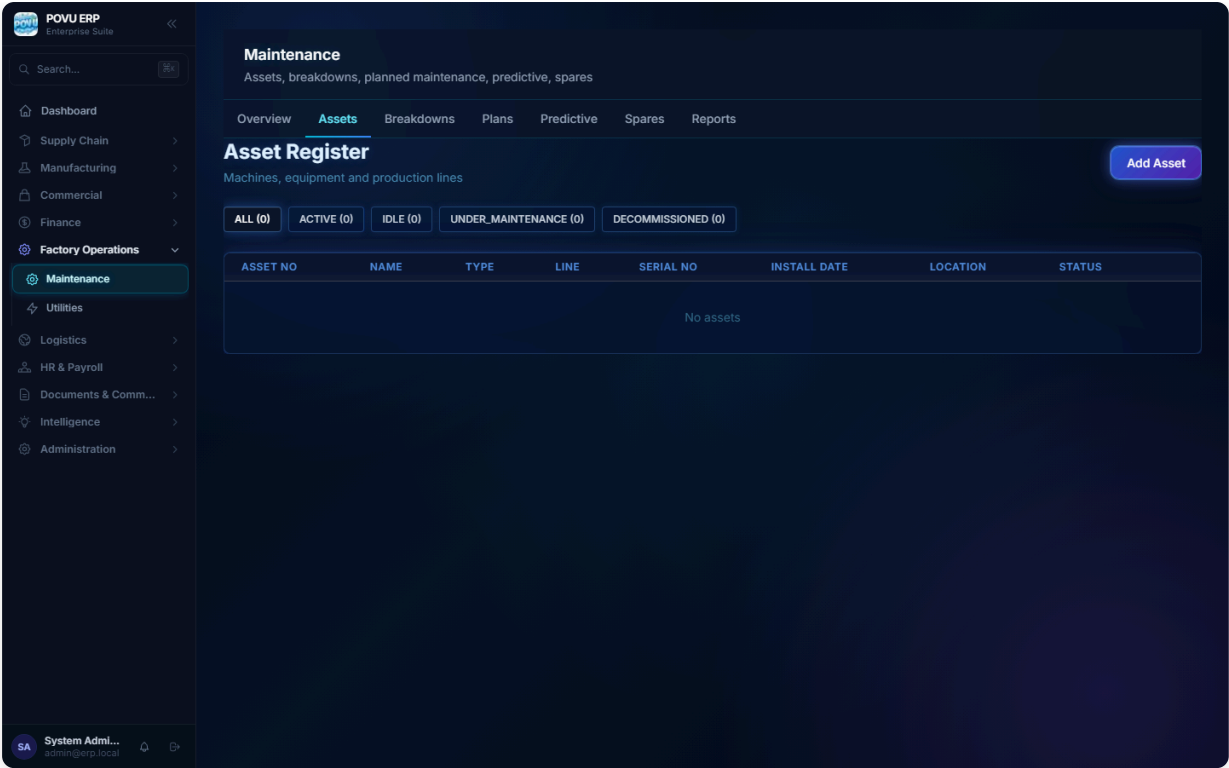
Charts: Breakdown frequency by machine, PM compliance rate, maintenance cost trend.

### 3. Asset Register

Tab: Assets

#### What it does

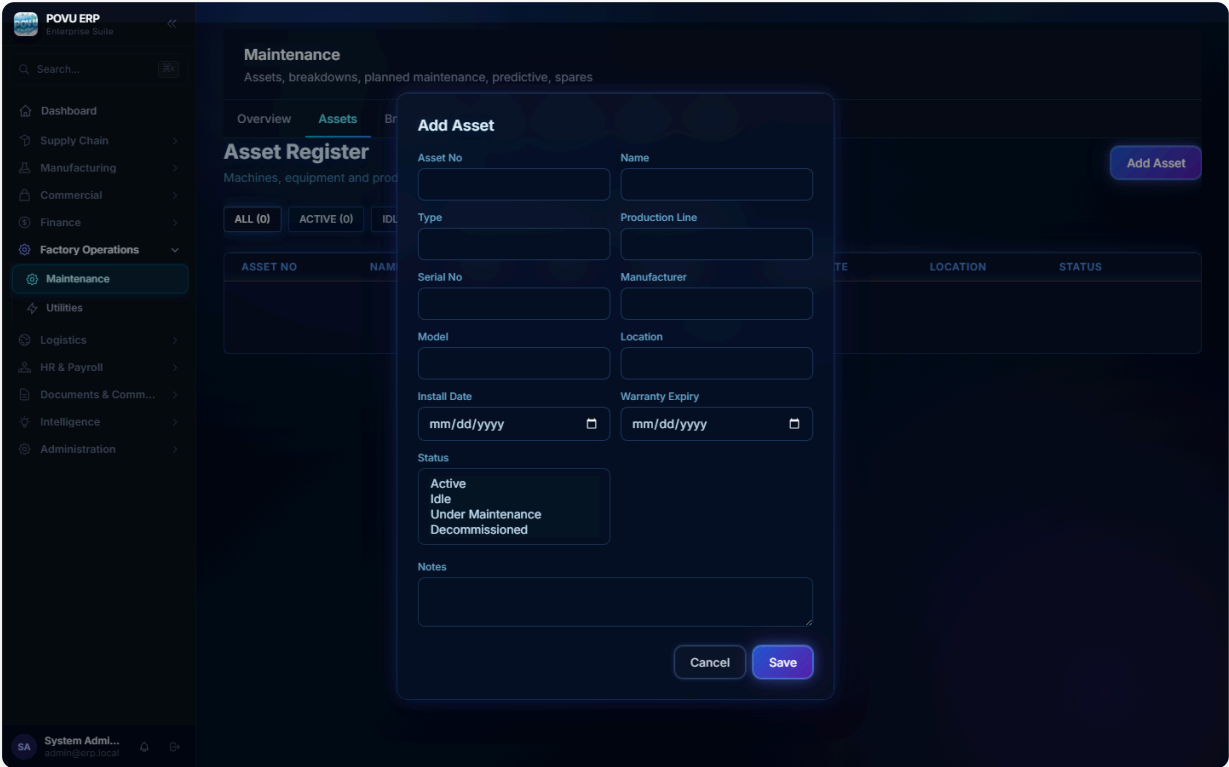
Maintains the master register of all equipment and facilities — machines, vehicles, utilities, and infrastructure.



Asset register — all assets with ID, name, location, status, and last maintenance date.

Creating a New Asset

Click + New Asset:



| Field               | Required | Notes                                       |
|---------------------|----------|---------------------------------------------|
| Asset Name          | Yes      | e.g. "Packaging Line 3 Filler"              |
| Asset Code          | Yes      | Unique identifier e.g. "PL3-FILL-001"       |
| Category            | Yes      | Machine / Vehicle / Infrastructure / IT     |
| Location            | Yes      | Factory area / line                         |
| Manufacturer        | No       | Equipment manufacturer                      |
| Model               | No       | Model number                                |
| Serial Number       | No       | Manufacturer serial number                  |
| Purchase Date       | No       | Date acquired                               |
| Purchase Cost       | No       | Acquisition cost                            |
| Warranty Expiry     | No       | Warranty end date                           |
| Useful Life (years) | No       | Expected operational life                   |
| PM Frequency        | No       | Days between preventive maintenance         |
| Criticality         | Yes      | Critical / Essential / Standard             |
| Status              | Yes      | Active / Under Maintenance / Decommissioned |

## Asset Criticality

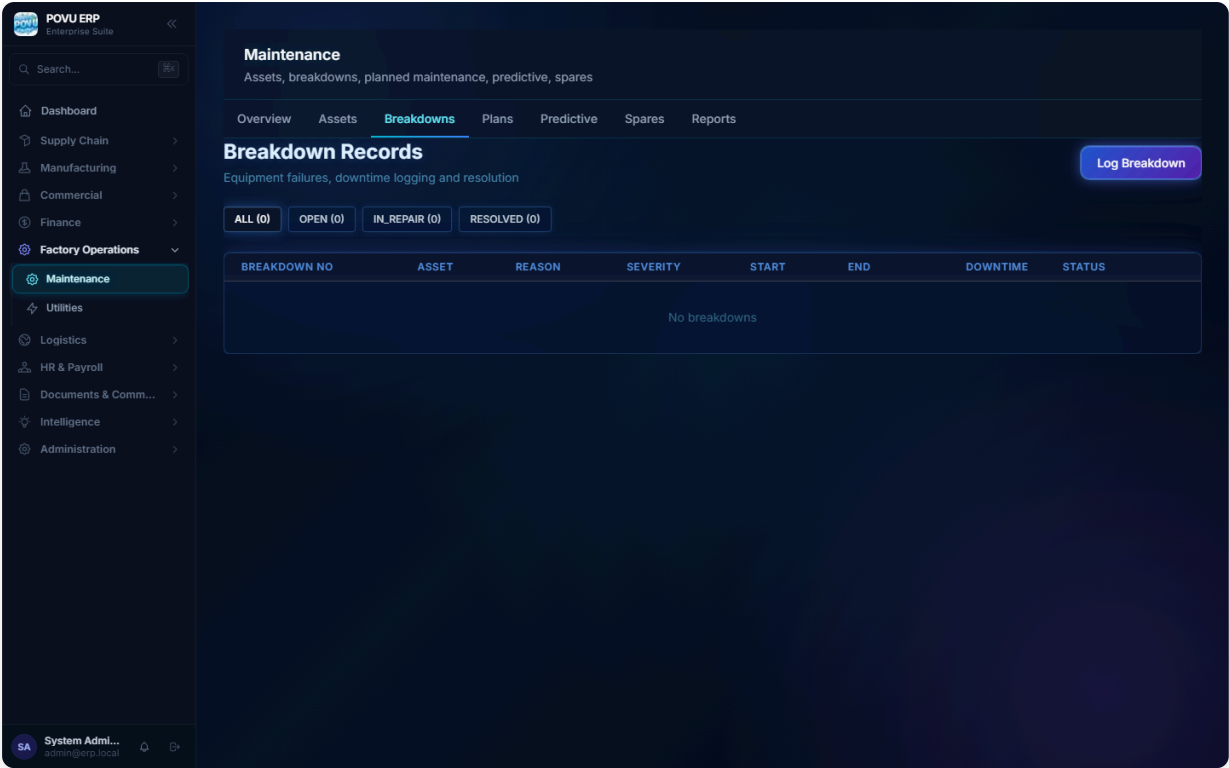
| Level     | Meaning                           | PM Priority            |
|-----------|-----------------------------------|------------------------|
| Critical  | Loss stops production entirely    | Highest; no deferral   |
| Essential | Loss significantly impacts output | High; 24-hour response |
| Standard  | Limited production impact         | Normal scheduling      |

## 4. Breakdown Management

**Tab:** Breakdowns

### What it does

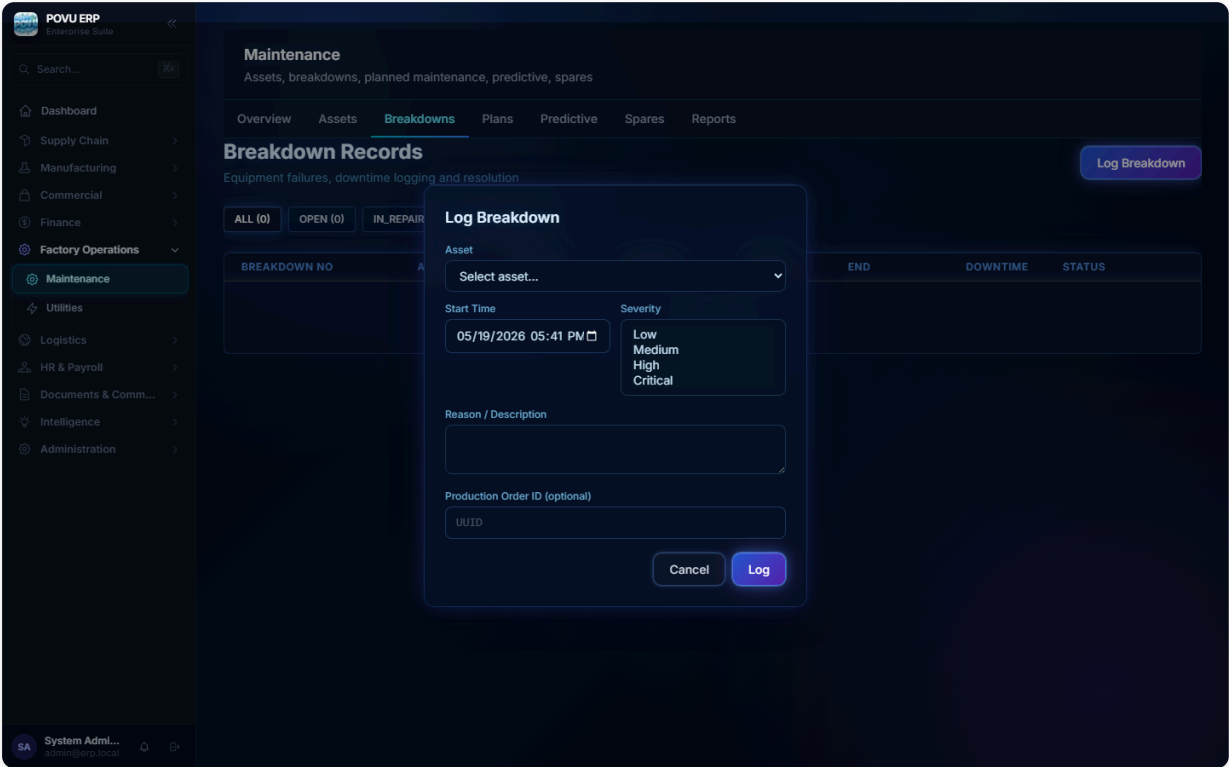
Log unplanned equipment failures — capture breakdown time, root cause, repair actions, parts used, and downtime duration.



Breakdown log — all breakdown events with machine, start time, resolution time, and root cause.

## Logging a Breakdown

Click + Log Breakdown:



| Field                 | Required | Notes                                                              |
|-----------------------|----------|--------------------------------------------------------------------|
| Asset                 | Yes      | Machine that failed                                                |
| Breakdown Start Time  | Yes      | Date and time failure occurred                                     |
| Failure Description   | Yes      | What happened                                                      |
| Failure Category      | Yes      | Mechanical / Electrical / Pneumatic / Hydraulic / Operator / Other |
| Severity              | Yes      | Minor / Major / Critical                                           |
| Reported By           | Yes      | Person logging the breakdown                                       |
| Assigned Technician   | No       | Who will repair                                                    |
| Estimated Repair Time | No       | Expected duration in hours                                         |

## Breakdown Workflow

1. **Log breakdown** → status: OPEN
2. Technician assigned → status: IN\_PROGRESS
3. **Log repair actions** — describe what was done
4. **Record parts used** — deduct from spare parts inventory
5. **Record actual repair time**
6. **Close breakdown** — enter root cause, corrective action, and preventive measure
7. Status: CLOSED → downtime minutes auto-calculated
8. Downtime fed to Production → OEE (if linked to machine)

## Breakdown Downtime to OEE

Closed breakdowns linked to a production machine automatically update OEE downtime records. Ensure Asset is linked to the correct Work Center in Manufacturing.

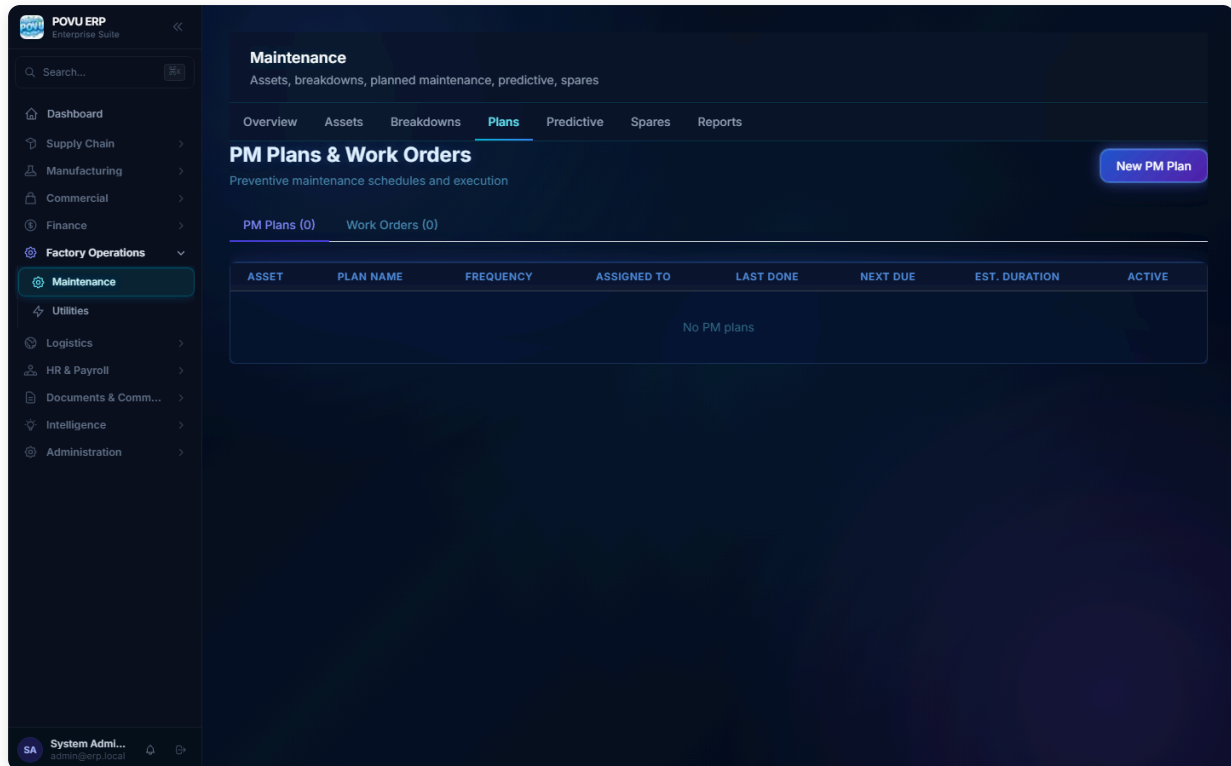
## 5. Planned Maintenance

**Tab:** Plans



## What it does

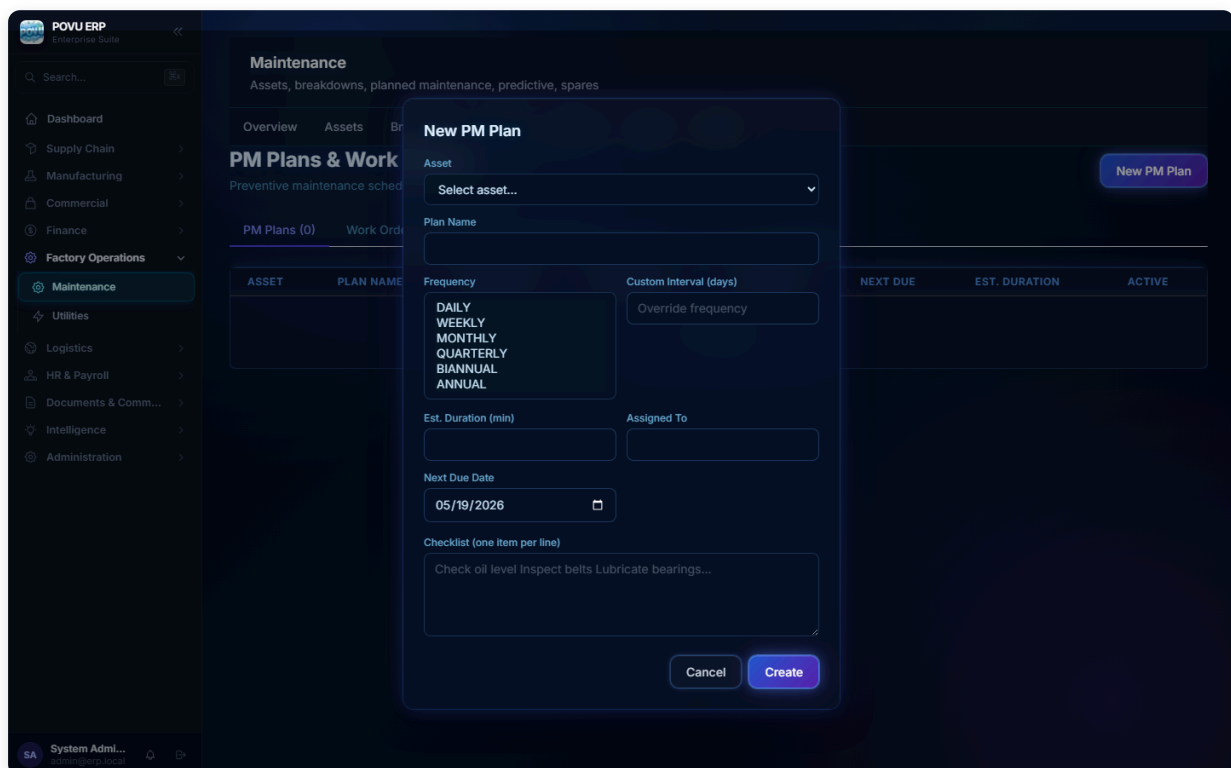
Schedule preventive maintenance tasks — periodic inspections, lubrication, calibration, and overhauls.



Planned maintenance list — upcoming PM tasks by asset and due date.

## Creating a Maintenance Plan

Click + New PM Plan:



| Field                    | Required | Notes                                                         |
|--------------------------|----------|---------------------------------------------------------------|
| Asset                    | Yes      | Target equipment                                              |
| Task Name                | Yes      | e.g. "Monthly Lubrication Check"                              |
| Task Type                | Yes      | Inspection / Lubrication / Calibration / Overhaul / Cleaning  |
| Frequency                | Yes      | Daily / Weekly / Monthly / Quarterly / Annual / Hours-based   |
| Interval                 | Yes      | Number (e.g. 30 for every 30 days or 500 for every 500 hours) |
| Estimated Duration (hrs) | Yes      | Time to complete                                              |
| Assigned To              | No       | Default technician                                            |
| Checklist                | No       | Step-by-step task checklist                                   |
| Next Due Date            | Yes      | First or next scheduled date                                  |
| Required Parts           | No       | Pre-stage spare parts                                         |

### PM Schedule View

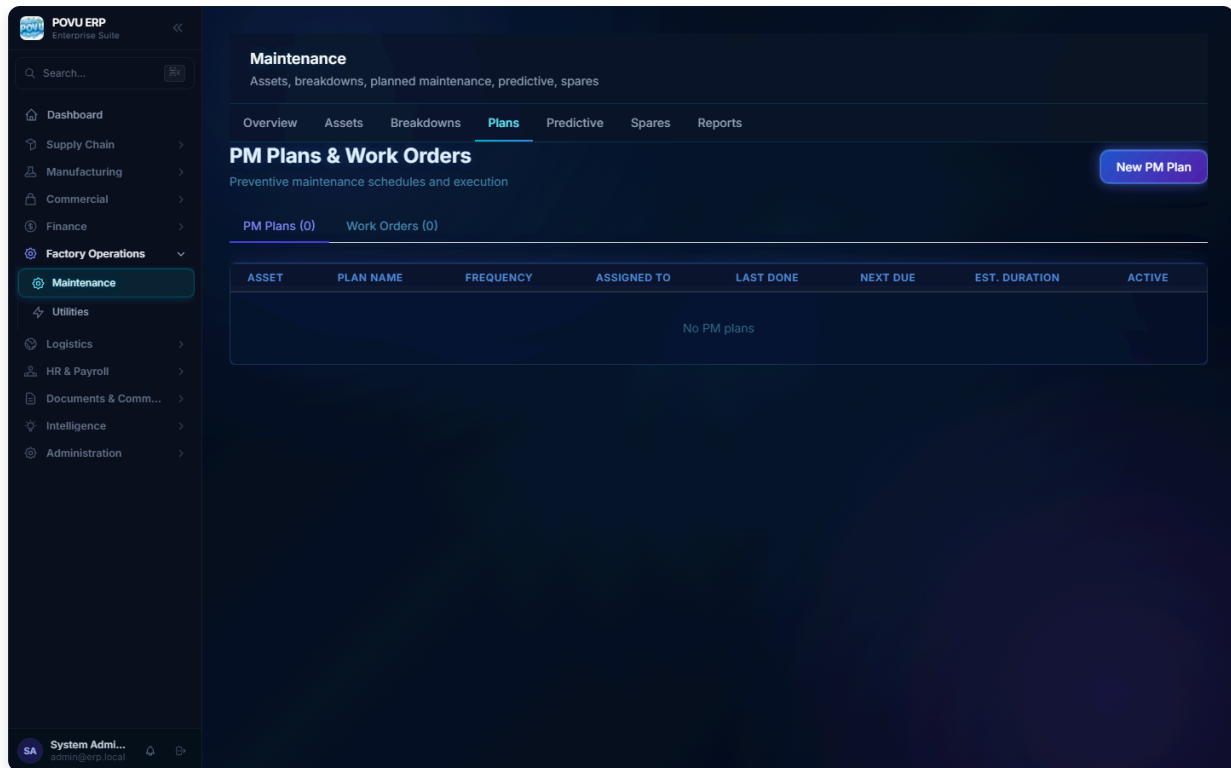
System auto-calculates next due dates. Upcoming tasks shown in calendar view. Overdue tasks flagged in red.

## 6. Maintenance Work Orders

**Tab:** Work Orders

### What it does

Formal work orders for both planned and breakdown maintenance activities.



*Maintenance work orders — all active and completed work orders.*

**Work Order Status Flow:** OPEN → ASSIGNED → IN\_PROGRESS → PARTS\_AWAITING → COMPLETED → CLOSED

**Work Order fields:**

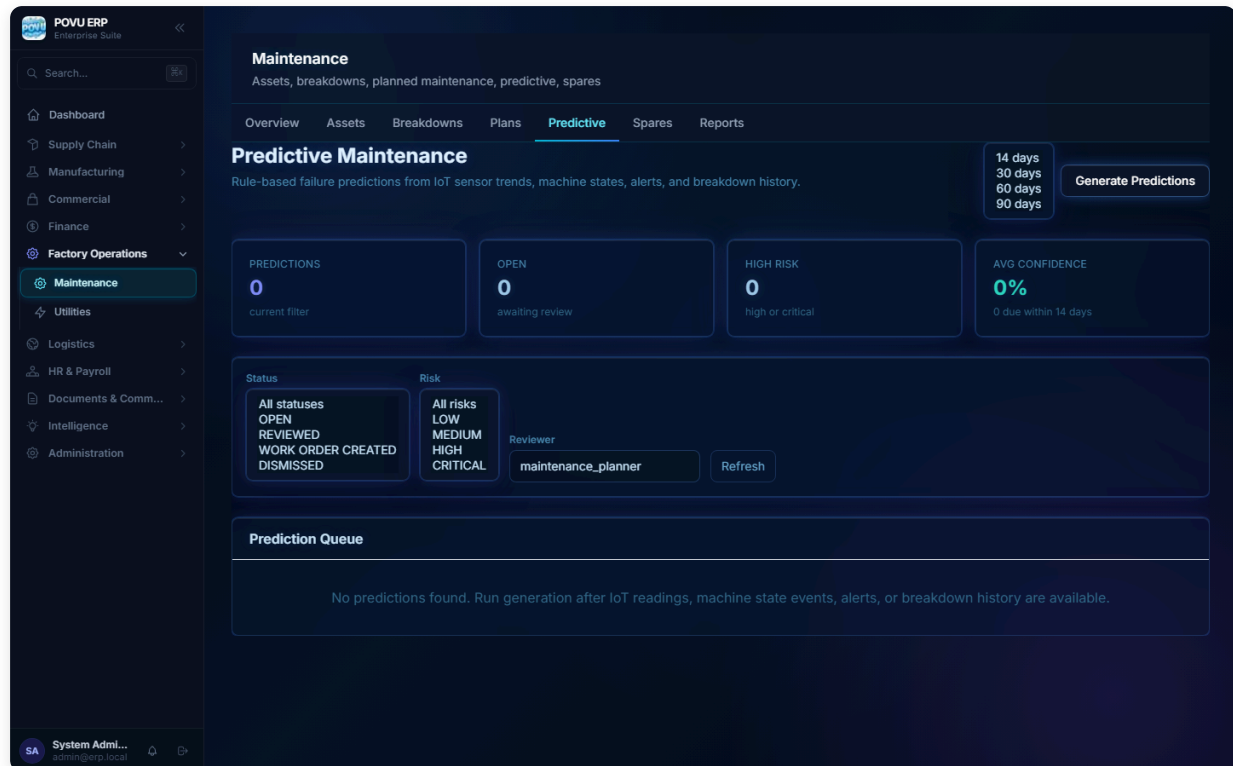
| Field               | Purpose                                       |
|---------------------|-----------------------------------------------|
| Asset               | Equipment being maintained                    |
| Type                | Breakdown / Preventive / Predictive / Project |
| Priority            | Emergency / High / Normal / Low               |
| Assigned Technician | Person doing the work                         |
| Scheduled Date      | When to perform                               |
| Actual Start / End  | For MTTR calculation                          |
| Parts Used          | Spare parts consumed                          |
| Labor Hours         | Time spent                                    |
| Root Cause          | For breakdown WOs                             |
| Work Performed      | Description of actions taken                  |

## 7. Predictive Maintenance

Tab: Predictive

### What it does

Monitors equipment condition data (vibration, temperature, oil analysis) to predict failures before they occur.



*Predictive maintenance — sensor readings, trend analysis, and risk alerts.*

### Condition monitoring parameters:

- Vibration (mm/s or g)
- Temperature (°C)
- Current draw (A)
- Oil viscosity/contamination
- Noise level (dB)

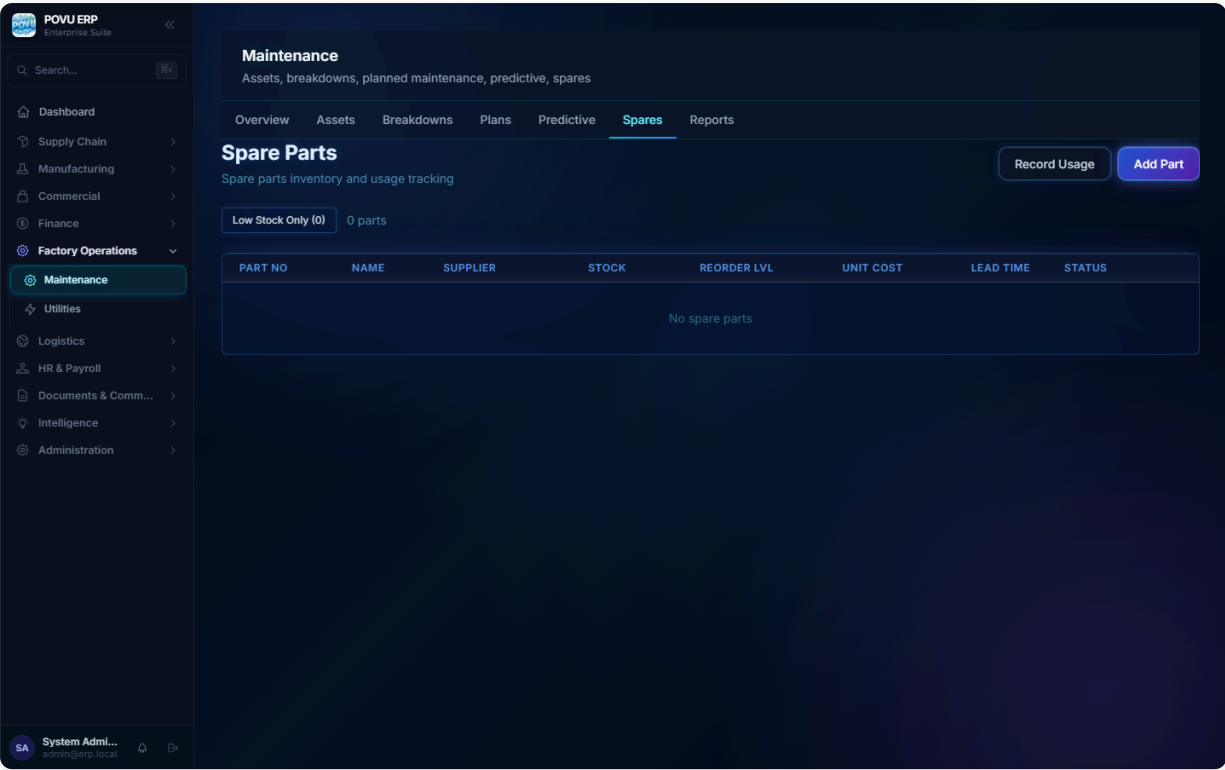
**Alerts:** System generates alert when reading exceeds threshold → creates predictive maintenance work order automatically.

## 8. Spare Parts Management

Tab: Spares

## What it does

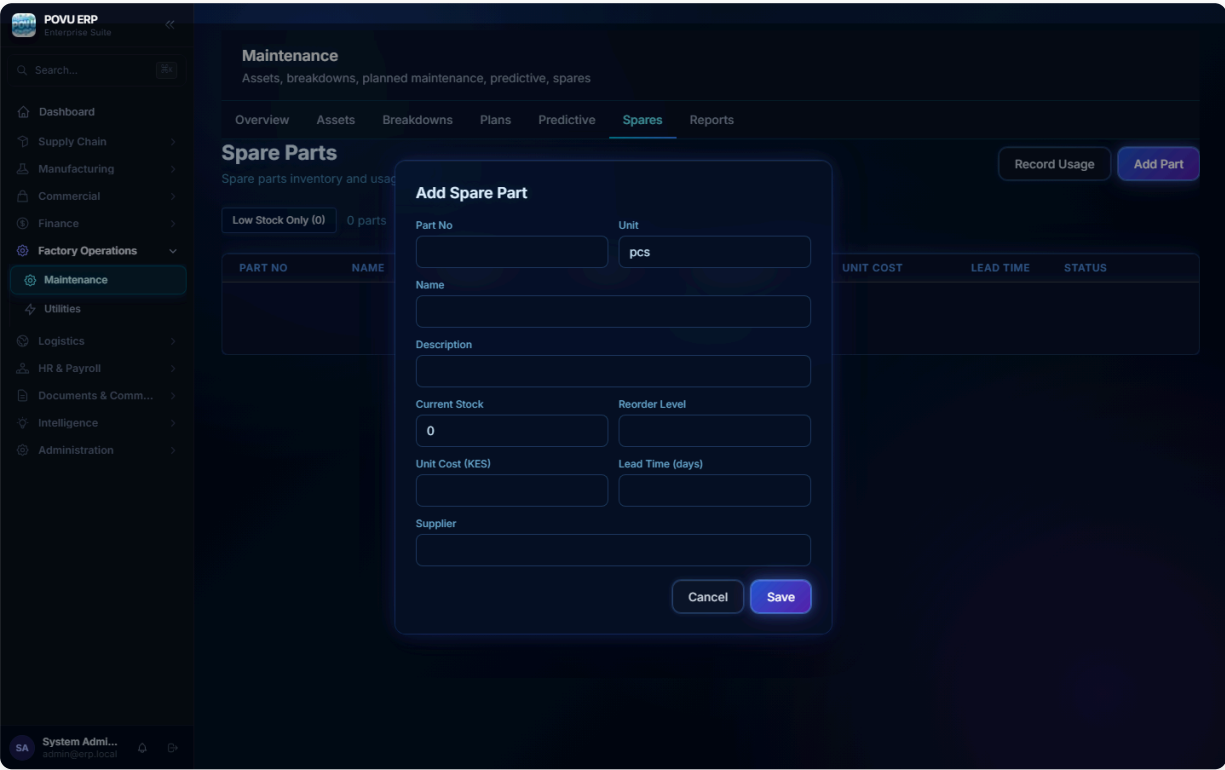
Manage spare parts inventory — stock levels, reorder points, part-to-asset linkage.



Spare parts register — part name, compatible assets, stock on hand, and reorder level.

## Adding a Spare Part

Click + Add Part:



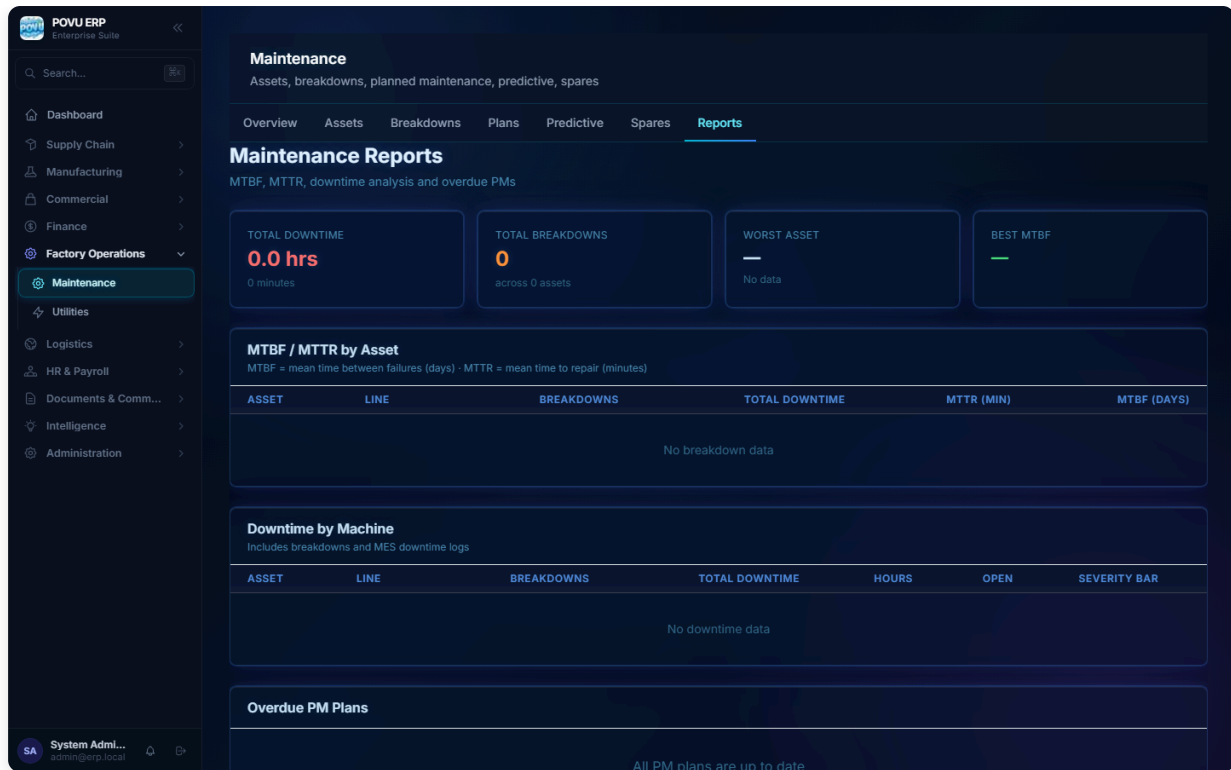
| Field             | Required | Notes                       |
|-------------------|----------|-----------------------------|
| Part Name         | Yes      | Descriptive name            |
| Part Number       | Yes      | Manufacturer part number    |
| Compatible Assets | No       | Assets this part is used on |
| Supplier          | No       | Preferred supplier          |
| Unit of Measure   | Yes      | Each / Kg / Litre           |
| Current Stock     | Yes      | Quantity on hand            |
| Reorder Point     | Yes      | Trigger level for reorder   |
| Reorder Quantity  | Yes      | Standard order quantity     |
| Unit Cost         | Yes      | For valuation               |
| Storage Location  | No       | Warehouse bin location      |

## Parts Usage

When a breakdown work order is closed with parts used, spare parts stock is automatically reduced. Low-stock alerts generated when stock falls below reorder point.

## 9. Maintenance Reports

**Tab:** Reports



Maintenance reports — MTBF/MTTR trends, breakdown frequency, PM compliance, cost analysis.

### Standard reports:

- MTBF by asset (Mean Time Between Failures)
- MTTR by asset (Mean Time To Repair)
- Breakdown frequency Pareto (which machines break most)
- PM compliance rate (% of planned maintenance completed on time)
- Maintenance cost by asset and category
- Spare parts consumption trend

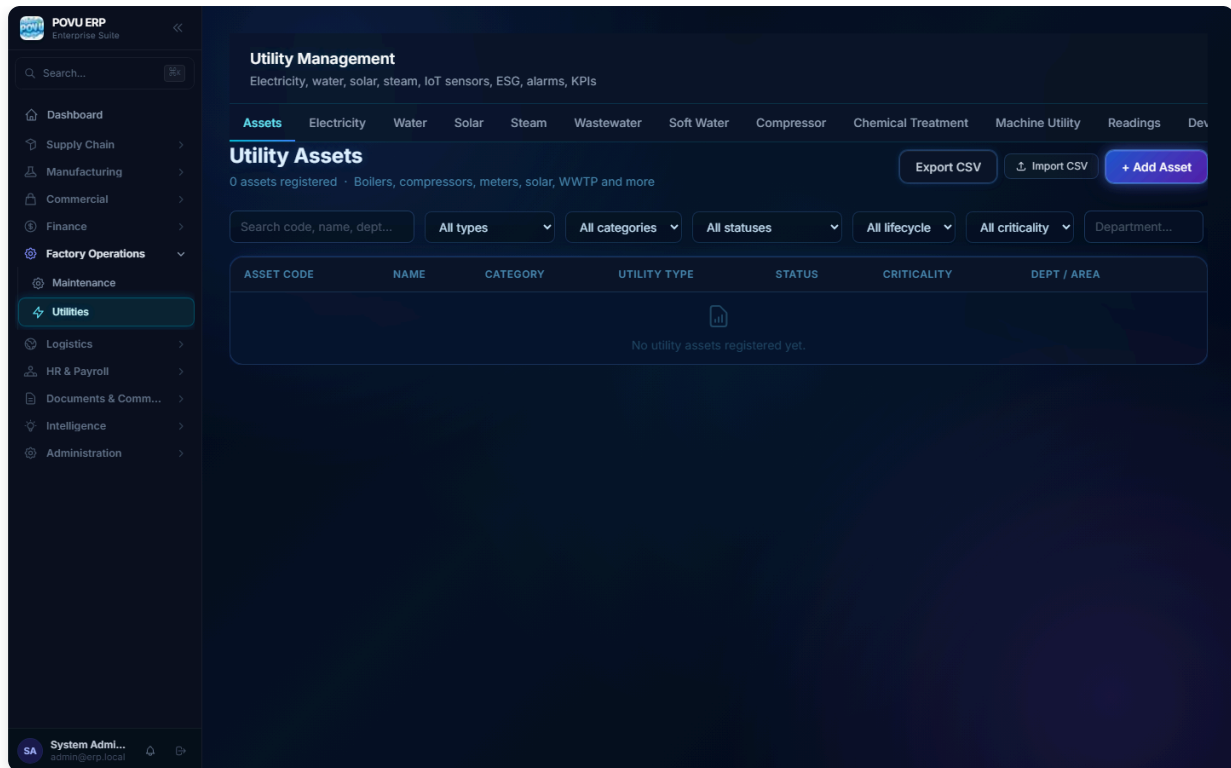
## 10. Utility Management — Electricity

Route: `/dashboard/utility-management`

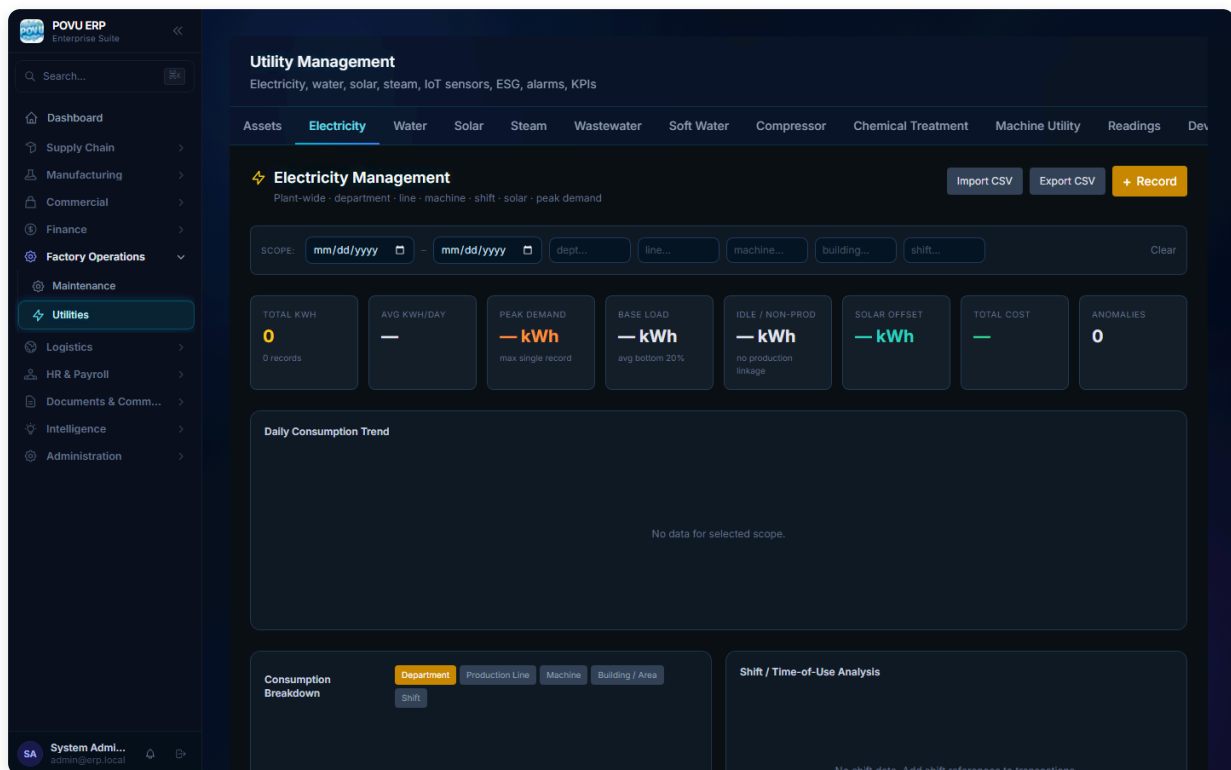
Tab: Electricity

### What it does

Track electricity consumption by meter, area, and machine. Monitor against targets and benchmarks.



Utility management overview.



Electricity consumption dashboard — kWh by area, peak demand, and cost.

## Electricity tracking:

- Meter readings entered daily/weekly
- System calculates consumption between readings
- Alerts on consumption exceeding threshold



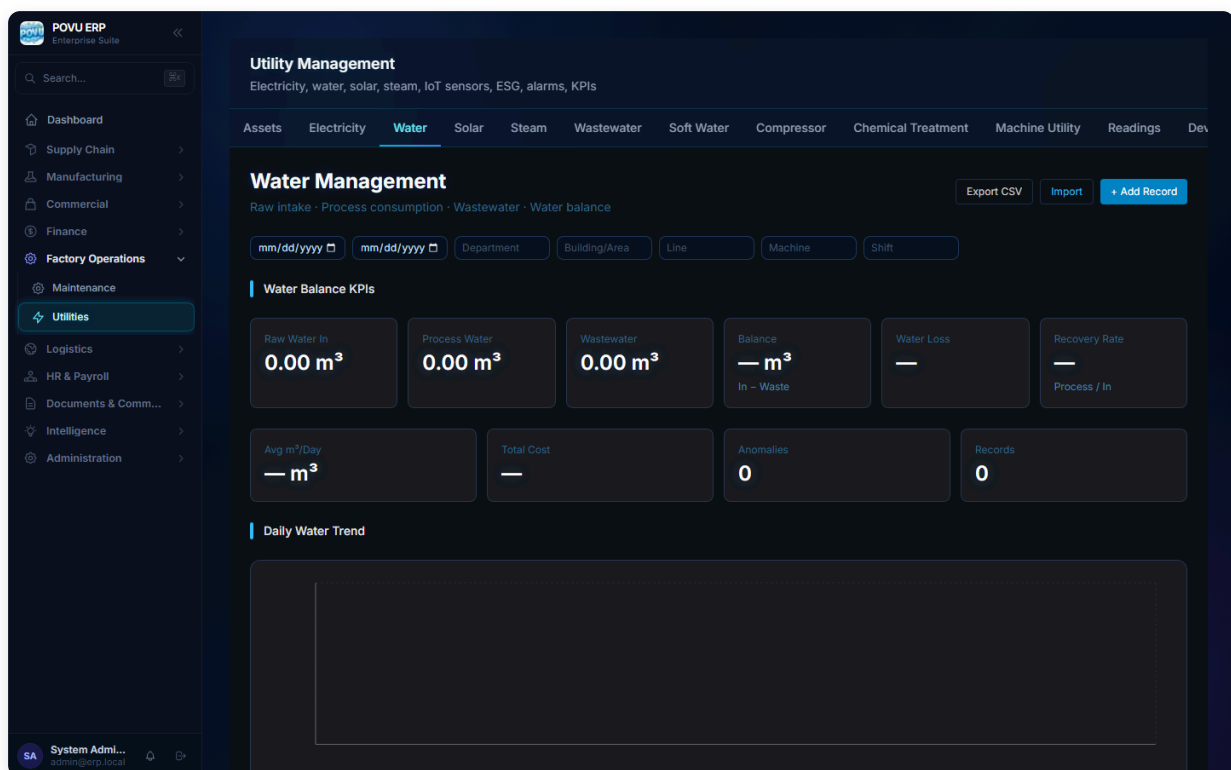
- Cost calculated using configured tariff rates
- Reports: consumption by area, by shift, by machine, cost per unit produced

## Recording a Meter Reading

1. Utility → Electricity → + New Reading
2. Select meter
3. Enter reading and date/time
4. System calculates kWh consumed since last reading

## 11. Utility Management — Water

Tab: Water



Water consumption dashboard —  $\text{m}^3$  by process area and cost.

Same workflow as electricity — meter readings, consumption calculation, and cost analysis.

### Water-specific tracking:

- Borehole extraction vs. municipal supply
- Process water vs. cleaning water separation
- Effluent discharge monitoring
- Water intensity (litres per unit produced)

## 12. Common Mistakes & Troubleshooting

| Problem                         | Cause                                         | Fix                                                            |
|---------------------------------|-----------------------------------------------|----------------------------------------------------------------|
| Breakdown not affecting OEE     | Asset not linked to work center               | Link asset to work center in Asset master                      |
| PM task not showing as overdue  | Frequency set incorrectly                     | Check PM plan interval and next due date                       |
| Spare parts stock not reducing  | Work order closed without entering parts used | Edit closed work order → add parts used → repost               |
| Predictive alert not triggering | Threshold not configured                      | Set alert thresholds in Predictive → Settings for each sensor  |
| Utility cost not calculating    | Tariff rate not set                           | Configure electricity/water tariff in Utility → Settings       |
| Breakdown root cause missing    | Work order closed without root cause          | Root cause is required for CLOSED status — reopen and complete |

## 13. Related Modules

| This Action         | Connects To                                      |
|---------------------|--------------------------------------------------|
| Breakdown logged    | Production → OEE (downtime record)               |
| PM work order       | Spare Parts (consumption) + HR (technician time) |
| Predictive alert    | Maintenance → Work Order (auto-created)          |
| Spare parts reorder | Procurement → Purchase Requisition               |
| Asset capitalized   | Finance → Fixed Assets                           |
| Utility cost        | Finance → Budgeting (utility expense)            |

*End of Maintenance & Utilities Manual v2*